

# MyDesign Portfolio

## *Element L*

### Element L: Presentation of Designer's Recommendations

- What could be done next to go further?  
What I would do next is to mass produce the light clamps for schools and tv studios around the globe. I would also make the light clamps more affordable and cost less resources to produce.
- What would you keep if you did the project again?  
I would keep the 3d printing aspect because that was very helpful to the creation of the light clamp and allowed us to bring our ideas to life.
- What would you change?  
I would change the clamping mechanism to be physically welded to the light holder in order for it to be more secure.
- What would have made this project easier for you to complete?  
I would probably want to have a more scheduled routine when we were going through the engineering design process.

In order to earn 5's on Element L, you will need specific and detailed ways the project can be improved and a rationale for why they are needed.

### Detailed Plan for Implementation of Improvements (L2)

#### 1. Mass Produce the Light Clamps for Schools and TV Studios

##### How to do it:

- Finalize the design so it's strong, safe, and easy to use.
- Work with manufacturers who can make large numbers affordably.
- Test a small batch in real classrooms or studios to make sure they work well.

- Sell them through school supply companies, studio gear websites, or online stores like Amazon.
- Make short videos or guides showing how useful and easy the clamp is.

**Why:**

Getting these clamps into more schools and studios can help people everywhere have better lighting setups without breaking the bank.

## 2. Make the Clamps More Affordable and Use Fewer Resources

**How to do it:**

- Use cheaper, eco-friendly materials like recycled plastic.
- Switch from 3D printing (for prototyping) to injection molding (for mass production) to lower costs.
- Simplify the design so fewer pieces are needed.
- Order parts in bulk to save money.
- Check that the production process isn't wasting materials or energy.

**Why:**

Lowering the cost and using fewer resources makes it easier for more people to afford the clamp—and it's better for the planet too.

## 3. Keep Using 3D Printing for Prototypes and Custom Orders

**How to do it:**

- Save your design files so they can be edited or improved in the future.
- Keep using 3D printers to try out new ideas quickly.
- Let schools download and print the clamp as a classroom project, so students can learn by building.

**Why:**

3D printing is great for testing ideas and teaching. It helps you move fast and get creative.

#### 4. Change the Clamp to Be Welded to the Light Holder

**How to do it:**

- Redesign the clamp so the connection is welded or built in, not just snapped on.
- Make sure the materials work well together and stay safe.
- Test the new welded version to make sure it's stronger and lasts longer.
- Update the production plan to include this new welded design.

**Why:**

A welded connection makes the clamp sturdier and more reliable—so lights stay put and don't fall.

#### 5. Improve the Project Schedule and Planning

**How to do it:**

- Create a simple timeline that shows what needs to be done and when.
- Have weekly check-ins to stay on track and talk through any problems.
- Give each person a clear role—like design lead, tester, or organizer.
- Break the work into smaller goals or “checkpoints.”
- Be flexible, but stay organized using tools like Google Calendar or Trello.

**Why:**

Better planning would make the whole process smoother, reduce last-minute stress, and help the team work better together.